MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TERMINATOR, RESISTOR-CAPACITOR, LINEAR, KEY CODE JBD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic resistor-capacitor terminator module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.86 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be J8D.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/14-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

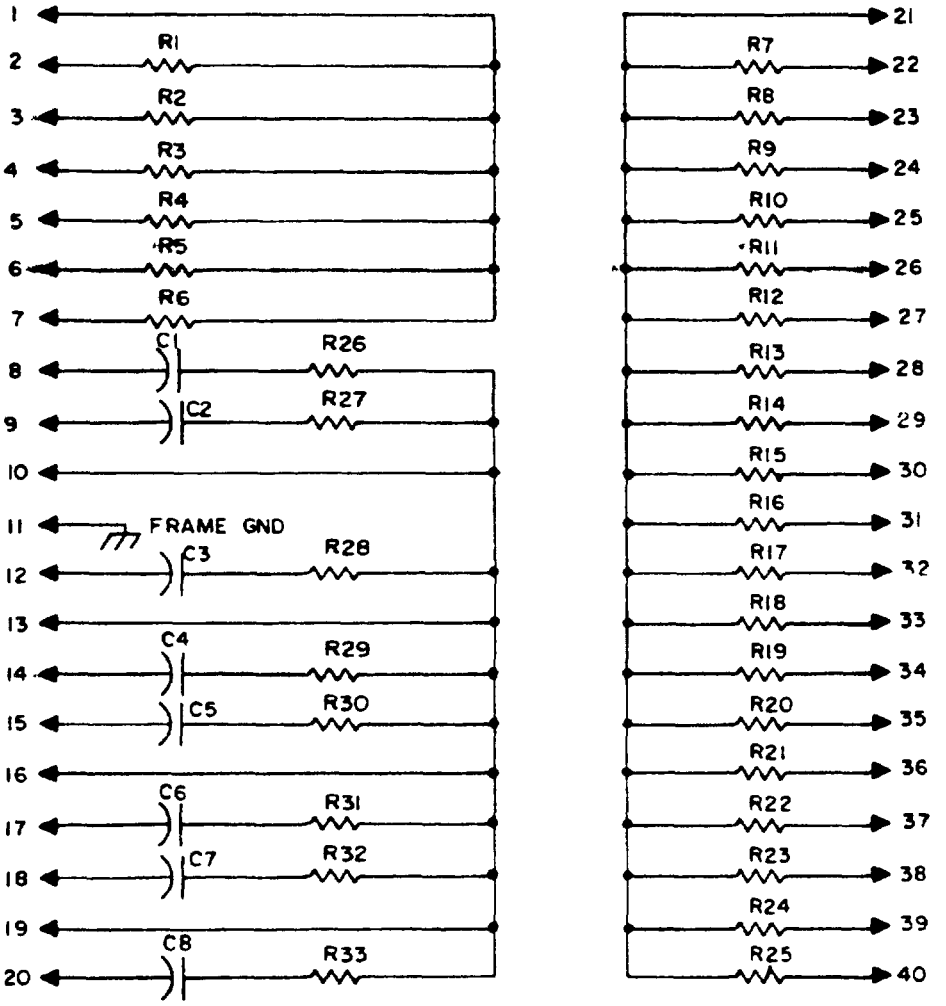
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES.
1. R1-R25 = 4.64 kΩ ±10%, 1/20 W.
2. R26-R33 = 100Ω ±2%, 1/20 W
3. C1-C8 = 330 pF ±20%, 100 V.

FIGURE 1. JBD module function diagram, terminator, resistor-capacitor.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	Integrity	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B8	Capacitor insulation resistance	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
C1-C25	Resistance	IV	X	4180	5100	4180	5100	4180	5100	4180	5100	Ω
D1-D3	Continuity	V	X	0	1	0	1	0	1	0	1	Ω
D4	Continuity	V	X	0	1	0	1	---	---	---	---	Ω
D5-D7	Continuity	V	Q	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	Ω
E1-E8	Output signal (10 MHz)	VI	X	0.94	1.22	0.92	1.24	0.93	1.23	0.92	1.24	Vp-p
F1-F8	Output signal (1 MHz)	VII	X	1.79	1.97	1.77	1.99	1.78	1.98	1.76	2.00	Vp-p

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through F shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through F in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C, D1 through D3, D5, through D7, E, and F, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be nonoperating. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through F shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through F shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

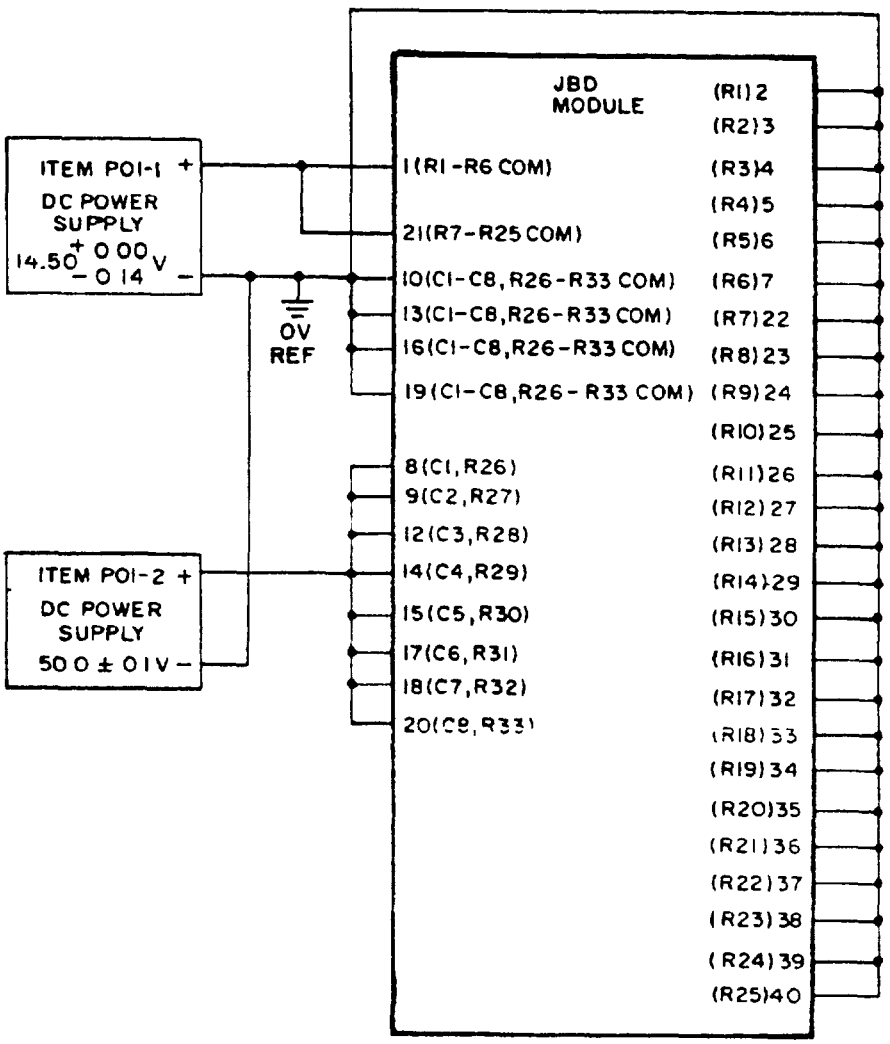
6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/14 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658731. Part number M28787/14-1 supersedes part numbers 2658731 and M28787/14.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	1.65 W
Resistor test voltage	14.5 V dc $^{+0\%}_{-1\%}$
Terminator test voltages	2 V_{p-p} ±3% @ 1 MHz ±5% 2 V_{p-p} ±3% @ 10 MHz ±5% 100 V dc ±10%

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* FIGURE 2 Operating temperature, component temperature, and life test circuit

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TABLE II. Integrity test.

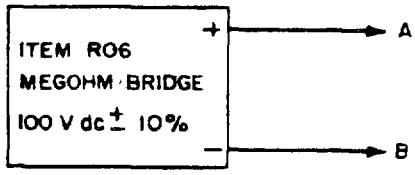
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Integrity	All pins except pin 11	11	10	∞	MΩ	3
A2	"	10	1	"	"	"	"
A3	"	21	1	"	"	"	"
A4	"	21	10	"	"	"	"
A5	"	1,9,12,14,15,17,18,20,21	8	"	"	"	"
A6	"	1,12,14,15,17,18,20,21	9	"	"	"	"
A7	"	1,14,15,17,18,20,21	12	"	"	"	"
A8	"	1,15,17,18,20,21	14	"	"	"	"
A9	"	1,17,18,20,21	15	"	"	"	"
A10	"	1,18,20,21	17	"	"	"	"
A11	"	1,20,21	18	"	"	"	"
A12	"	1,21	20	"	"	"	"

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TABLE III. Capacitor insulation resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Capacitor insulation resistance	8	10	10	∞	MΩ	3
B2	"	9	"	"	"	"	"
B3	"	12	"	"	"	"	"
B4	"	14	"	"	"	"	"
B5	"	15	"	"	"	"	"
B6	"	17	"	"	"	"	"
B7	"	18	"	"	"	"	"
B8	"	20	"	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

JBD
MODULE

1(R1-R6 COM)	(R7-R25 COM)21
2(R1)	(R7)22
3(R2)	(R8)23
4(R3)	(R9)24
5(R4)	(R10)25
6(R5)	(R11)26
7(R6)	(R12)27
8(C1,R26)	(R13)28
9(C2,R27)	(R14)29
10(C1-C8,R26-R33 COM)	(R15)30
11 FRAME GND	(R16)31
12(C3,R28)	(R17)32
13(C1-C8,R26-R33 COM)	(R18)33
14(C4,R29)	(R19)34
15(C5,R30)	(R20)35
16(C1-C8,R26-R33 COM)	(R21)36
17(C6,R31)	(R22)37
18(C7,R32)	(R23)38
19(C1-C8,R26-R33 COM)	(R24)39
20(C8,R33)	(R25)40

NOTE For test conditions B1-B8, the bridge voltage shall be applied between the module pins for a minimum of two minutes prior to making the measurement.

FIGURE 3. Integrity and capacitor insulation resistance test circuit for tables II and III.

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TABLE IV. Resistance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	R1	Resistance	2	1	4180	5100	Ω	4
C2	R2	"	3	"	"	"	"	"
C3	R3	"	4	"	"	"	"	"
C4	R4	"	5	"	"	"	"	"
C5	R5	"	6	"	"	"	"	"
C6	R6	"	7	"	"	"	"	"
C7	R7	"	22	21	"	"	"	"
C8	R8	"	23	"	"	"	"	"
C9	R9	"	24	"	"	"	"	"
C10	R10	"	25	"	"	"	"	"
C11	R11	"	26	"	"	"	"	"
C12	R12	"	27	"	"	"	"	"
C13	R13	"	28	"	"	"	"	"
C14	R14	"	29	"	"	"	"	"
C15	R15	"	30	"	"	"	"	"
C16	R16	"	31	"	"	"	"	"
C17	R17	"	32	"	"	"	"	"
- - -								

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TABLE IV. Resistance test - Continued.

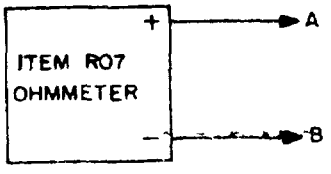
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C18	R18	Resistance	33	21	4180	5100	Ω	4
C19	R19	"	34	"	"	"	"	"
C20	R20	"	35	"	"	"	"	"
C21	R21	"	36	"	"	"	"	"
C22	R22	"	37	"	"	"	"	"
C23	R23	"	38	"	"	"	"	"
C24	R24	"	39	"	"	"	"	"
C25	R25	"	40	"	"	"	"	"

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* TABLE V. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Continuity	13	10	0	1	Ω	4
D2	"	16	"	"	"	"	"
D3	"	19	"	"	"	"	"
D4	"	11	Frame	"	"	"	"
D5	"	13	10	0.0	0.5	"	"
D6	"	16	"	"	"	"	"
D7	"	19	"	"	"	"	"

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NOTE For test condition D4, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

JBD MODULE	
1(R1-R6 COM)	(R7-R25 COM)21
2(R1)	(R7)22
3(R2)	(R8)23
4(R3)	(R9)24
5(R4)	(R10)25
6(R5)	(R11)26
7(R6)	(R12)27
8(C1,R26)	(R13)28
9(C2,R27)	(R14)29
10(C1-C8,R26-R33 COM)	(R15)30
11 FRAME GND	(R16)31
12(C3,R28)	(R17)32
13(C1-C8,R26-R33 COM)	(R18)33
14(C4,R29)	(R19)34
15(C5,R30)	(R20)35
16(C1-C8,R26-R33 COM)	(R21)36
17(C6,R31)	(R22)37
18(C7,R32)	(R23)38
19(C1-C8,R26-R33 COM)	(R24)39
20(C8,R33)	(R25)40

FIGURE 4. Resistance and continuity test circuit for tables IV and V.

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TABLE VI. Output signal (10 MHz) test.

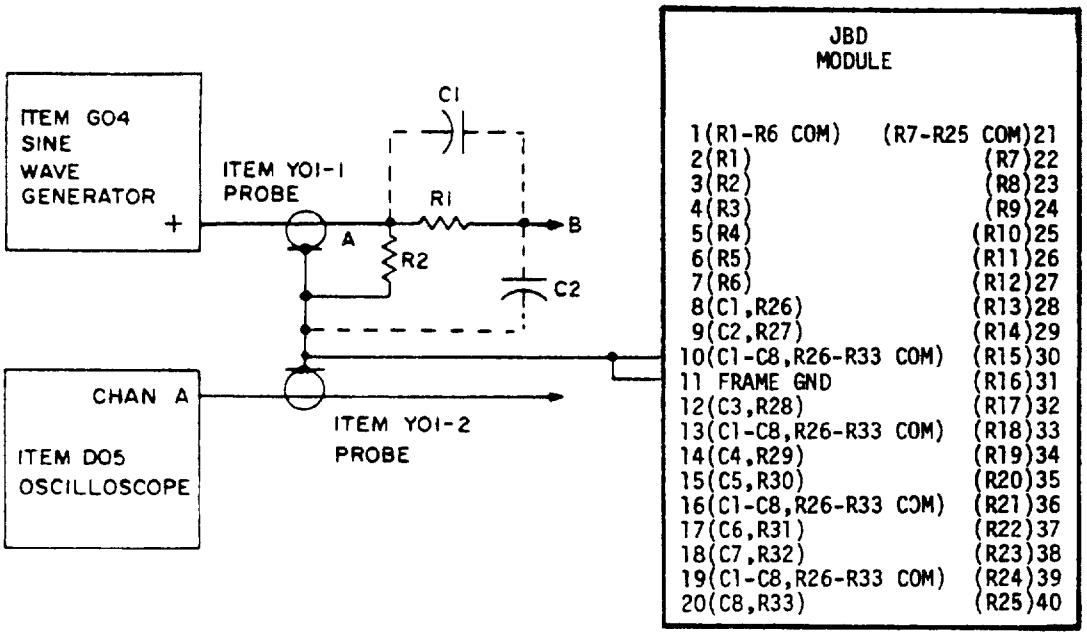
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		B to pin	Min	Max		
E1	Output signal (10 MHz)	8	0.94	1.22	V_{p-p}	5
E2	"	9	"	"	"	"
E3	"	12	"	"	"	"
E4	"	14	"	"	"	"
E5	"	15	"	"	"	"
E6	"	17	"	"	"	"
E7	"	18	"	"	"	"
E8	"	20	"	"	"	"

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TABLE VII. Output signal (1 MHz) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		B to pin	Min	Max		
F1	Output signal (1 MHz)	8	1.79	1.97	V_{p-p}	5
F2	"	9	"	"	"	"
F3	"	12	"	"	"	"
F4	"	14	"	"	"	"
F5	"	15	"	"	"	"
F6	"	17	"	"	"	"
F7	"	18	"	"	"	"
F8	"	20	"	"	"	"

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NOTES

1 R1 = 100Ω ±1%, 1/4 W

2 R2 = 50Ω ±2%, 1 W

3 C1 = 10 pF maximum (including probe and parasitic capacitance measured at the pin under test at 10 MHz with the module removed from the socket)

4 C2 = 12 pF maximum (including probe and parasitic capacitance measured at the pin under test at 10 MHz with the module removed from the socket)

5 The maximum allowable wire length between the following points shall be 2 inches (50.8 mm).

a Point B to module pins 8, 9, 12, 14, 15, 17, 18 and 20
b Point A to point B
c Items Y01-1 and Y01-2 ground point to pins 10 and 11 of the module under test.

6 Test sequence for table VI

a Connect point B to the module pin specified in table VI, test condition E1.
b. With item Y01-2 connected to point A, adjust item G04 for a sine wave period of 0.1 µs ±5% and an amplitude of 2 V_{pp} ±3% as read on item D05.
c Using item Y01-2, measure V_{pp} at point B. The measured value shall be as specified in table VI, test condition E1.
d Repeat notes 6a thru 6c for test condition E2 thru E8 of table VI.

7. Test sequence for table VII

a. Connect point B to the module pin specified in table VII, test condition F1.
b With item Y01-2 connected to point A, adjust item G04 for a sine wave period of 1 µs ±5% and an amplitude of 2 V_{pp} ±3% as read on item D05
c Using item Y01-2, measure V_{pp} at point B. The measured value shall be as specified in table VII, test condition F1
d Repeat notes 7a thru 7c for test condition F2 thru F8 of table VII.

FIGURE 5 Output signal (10 MHz and 1 MHz) test circuit for tables VI and VII.

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* 6.5 Module function. This module contains eight resistor-capacitor networks consisting of 100Ω ±2 percent, 1/20 W, resistors in series with 330 pF ±20 percent, 100 V, capacitors. The module also contains 25 each 4.64 kΩ ±10 percent, 1/20 W, pull-up resistors.

6.5.1 Terminating circuits. Each terminating circuit is an ac terminator and presents no dc loading to logic gates or power supply. Output lines driven by DTL NAND gates and monostable multivibrators can be terminated with this module to minimize ringing. Additional ground pins are provided to facilitate making twisted pair connections.

6.5.2 Logic circuit pull-up resistors The 4.64 kΩ resistors may be used as logic circuit pull-up resistors.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/14(ER)	MODULES, STANDARD ELECTRONIC TERMINATOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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